

# Induced potential, transformers and the National Grid

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 13 starred (★) questions first.

|         | QUESTION                                                                   | ANSWER                                                                                                  | ✓                        |
|---------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------|
| 1<br>★  | State what is induced by the generator effect.                             | A potential difference (and a current, in a complete circuit).                                          | <input type="checkbox"/> |
| 2<br>★  | State what is needed for the generator effect to occur.                    | A conductor moving relative to a magnetic field, or a changing magnetic field around a conductor.       | <input type="checkbox"/> |
| 3<br>★  | Give three ways to increase the size of the induced potential difference.  | Use a stronger magnetic field, move faster, or use more turns on the coil.                              | <input type="checkbox"/> |
| 4       | Describe how to reverse the direction of the induced potential difference. | Reverse the direction of movement or of the magnetic field.                                             | <input type="checkbox"/> |
| 5<br>★  | State the difference between an alternator and a dynamo.                   | An alternator generates alternating current; a dynamo generates direct current.                         | <input type="checkbox"/> |
| 6       | Name the effect used in microphones.                                       | The generator effect                                                                                    | <input type="checkbox"/> |
| 7<br>★  | Describe how a microphone works.                                           | Sound makes a diaphragm and coil vibrate in a magnetic field, inducing a changing potential difference. | <input type="checkbox"/> |
| 8       | Name the effect used in transformers.                                      | The generator effect                                                                                    | <input type="checkbox"/> |
| 9<br>★  | Describe the structure of a transformer.                                   | A primary coil and a secondary coil wound on an iron core.                                              | <input type="checkbox"/> |
| 10      | Describe what a step-up transformer does.                                  | It increases the potential difference.                                                                  | <input type="checkbox"/> |
| 11      | Describe what a step-down transformer does.                                | It decreases the potential difference.                                                                  | <input type="checkbox"/> |
| 12<br>★ | Compare the numbers of turns on the coils of a step-up transformer.        | More turns on the secondary coil than on the primary.                                                   | <input type="checkbox"/> |

| QUESTION |                                                                                             | ANSWER                                                                                                                   | ✓                        |
|----------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 13       | Compare the numbers of turns on the coils of a step-down transformer.                       | Fewer turns on the secondary coil than on the primary.                                                                   | <input type="checkbox"/> |
| 14<br>★  | State how power input and power output are related in an ideal transformer.                 | They are equal.                                                                                                          | <input type="checkbox"/> |
| 15       | Name the type of current supplied to the primary coil of a transformer.                     | Alternating current                                                                                                      | <input type="checkbox"/> |
| 16       | Name the type of current induced in the secondary coil of a transformer.                    | Alternating current                                                                                                      | <input type="checkbox"/> |
| 17<br>★  | Explain why a transformer needs an iron core.                                               | Iron is easily magnetised, so it carries the changing magnetic field from the primary coil to the secondary coil.        | <input type="checkbox"/> |
| 18<br>★  | State what the magnetic field of an induced current does.                                   | It opposes the change that caused it.                                                                                    | <input type="checkbox"/> |
| 19<br>★  | Explain why a transformer does not work with a direct current.                              | Direct current gives a constant magnetic field; induction needs a changing field.                                        | <input type="checkbox"/> |
| 20       | State the equation linking the potential differences and numbers of turns of a transformer. | p.d. across primary ÷ p.d. across secondary = turns on primary ÷ turns on secondary<br>( $V_p \div V_s = n_p \div n_s$ ) | <input type="checkbox"/> |
| 21<br>★  | Describe the potential difference–time graph for an alternator.                             | It alternates: a wave going equally above and below zero.                                                                | <input type="checkbox"/> |
| 22<br>★  | Describe the potential difference–time graph for a dynamo.                                  | It rises and falls but never goes below zero.                                                                            | <input type="checkbox"/> |